

Compiler Design DSL Project Report

1. Project Description

This project implements a simple Domain Specific Language (DSL) compiler using Flex and Bison. The compiler processes input statements, performs lexical analysis, parsing, and generates output.

2. Architecture

The compiler follows the standard phases: Lexical Analysis → Parsing → Abstract Syntax Tree → Semantic Analysis → Intermediate Code Generation.

3. Modules Explanation

Lexical Analyzer: Identifies tokens using Flex.

Parser: Validates syntax using Bison.

AST: Represents structure of code.

Semantic Analysis: Checks meaning and correctness.

ICG: Generates intermediate representation.

4. Tools Used

C Programming Language, Flex (Lex), Bison (YACC), GitHub for version control.

5. Sample Input

```
x = 10;
```

6. Output

Valid statement

7. Conclusion

The project demonstrates the basic working of a compiler including parsing and validation of a simple language.